

Name of Student: Mritunjay PandeyScholar No.: 23VQ2078**INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL****Department of CSE****Mid-Term Examination September - 2024**

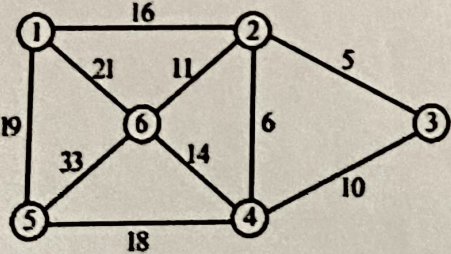
Subject Code: CSE-215

Subject Name: Design & Analysis of Algorithms

Time Allotted: 90 min

MM: 20

- NOTE: 1. All questions are compulsory.
2. Mark answer number clearly.

1. (a)	Define algorithm. Explain asymptotic notations Big O, big Omega, big theta notations. Prove that if $g(n)$ is $\Omega(f(n))$ then $f(n)$ is $O(g(n))$.	3
(b)	What is the time complexity of "find(n)"? <pre>find(int n) { if (n < 2) then return n; else { int sum= 0; for (int i= 1; i ≤ 4; i++) find(n /2); for (int i=1; i≤ n*n; i++) sum= sum + 1; } }</pre> Assume that the division operation takes constant time and "sum" is global variable	2
2.	Formulate the Knapsack problem with greedy method and find the optimal solution for $n=7$, $m=15$, $(p_1-p_7)=(10,5,15,7,6,18,3)$, $(w_1-w_7)=(2,3,5,7,1,4,1)$.	5
3:	What is the time complexity to execute the kruskal's algorithm (To find the minimum spanning tree) on the following graph. State the time taken by the algorithm at each step. 	5
4. (a)	Describe the importance of pivot element in quick sort, How the position of it affects the performance of quick sort? Explain with algorithms.	3
(b)	Write a Huffman coding algorithm. Construct the Huffman code for the given	2

frequencies: F:10, E:19, C:21, B:23, D:36, A:45